Mithilesh Vaidya

vaidya.mithilesh@gmail.com ▪ methi1999.github.io ▪ LinkedIn

Education

- **Georgia Institute of Technology**, Atlanta, USA [2022 - 2024]
  - Master's in Computer Science (Specialisation: Machine Learning)

- **Indian Institute of Technology Bombay**, Mumbai, India [2017 - 2022]
  - Bachelor and Master of Technology in Electrical Engineering, **CPI 9.51/10**
  - **Minor** in Computer Science and Engineering

Honors and Awards

- **Institute Silver Medal** at IIT Bombay for best academic standing in department [2022]
- **Excellence in Academics** Award, for excellent academic performance (Top 3/121) [2022]
- **Undergraduate Research Award** (URA03) for outstanding thesis contributions [2022]
- Bhavesh Gandhi Memorial Prize for standing 1st in the CSP specialization
- **JN Tata** Scholarship for pursuing higher education abroad [2022]
- **AP grade** in **Control Systems** course for **exceptional performance** [2020]
- **All India Rank of 388** in JEE Advanced 2017 among 2,00,000 candidates [2017]
- **Kishore Vaigyanik Protsahan Yojana (KVPY)** Fellowship by Govt. of India [2015]
- **National Talent Search Examination (NTSE)** scholarship by Govt. of India [2015]
- **Silver medal** in Homi Bhabha Young Scientist Examination [2011]

Publications and Pre-prints

- A. Luebs, **Mithilesh Vaidya**, I. Kumar, S. Badam, S. W. Bailey, M. Bendel, J. Sotelo and X. He. "PoDAR: Power-Disentangled Audio Representation for Generative Modeling." arXiv preprint arXiv:2605.10084 (2026).
- **M. Vaidya**, C. Rodgers and A. Wu. "Switching Subspace Model for Neural Population Analysis" (2024).
- **Vaidya, Mithilesh**, Binaya Kumar Sahoo, and Preeti Rao. "Deep Learning for Assessment of Oral Reading Fluency." arXiv preprint arXiv:2405.19426 (2024).
- **M. Vaidya**, K. Sabu and P. Rao. "Deep Learning for Prominence Detection In Children's Read Speech." arXiv preprint arXiv:2110.14273 (2021).
- Sabu, Kamini, **Mithilesh Vaidya**, and Preeti Rao. "CNN Encoding of Acoustic Parameters for Prominence Detection." arXiv preprint arXiv:2104.05488 (2021).

Professional Experience

- **AI Researcher** *June'24 - Present*
  Descript, San Francisco *Research Engineer*
  - Co-led development of **Audio Regenerate**, a production latent-inpainting system for seamless word-level audio editing in Descript
  - Trained both stages of the Audio Regenerate pipeline: a continuous neural audio codec and a flow-matching generator for masked latent inpainting
  - Developed **Video Regenerate**, an audio-driven lip-sync model for video editing and translation, including a contrastive model for audio-video alignment
  - Co-authored **PoDAR**, a power-disentangled audio representation that improves latent modelability and accelerates generator convergence by $\mathbf{2}\times$
  - Improved the data preprocessing pipeline for training generative video models on large-scale datasets
- **LLM-powered Natural Language Interface** *May'23 - Aug'23*
  Qualcomm, San Diego | Guide: Mr. Vasudev Nayak, Principal Engineer *Machine Learning Internship*
  - Designed an end-to-end hierarchical modular pipeline for structured parsing using LLMs such as MPT-7B
  - Developed techniques for reducing hallucinations and compared performance across various open-source LLMs
  - Evaluated performance of NVIDIA Nemo and OpenAI Whisper for STT and machine translation

- **Verification of FPGA-based High Frequency Trading Platform** *Apr'20 - June'20*

APT Portfolio Pvt. Ltd. | Guide: Mr. Vivek Pannikar, Senior Verification Engineer *Internship*

- Implemented **DPI**, a protocol for exchanging data between SystemVerilog and other languages, for **speeding up verification** of testbenches using Cocotb, Quartus and Riviera by **3x**
- Used Python **metaclasses** for automatically generating Python, SystemVerilog and C DPI header and implementation files from high-level JSON inputs

- **Autonise AI** *Sep'18 - May'19*
  Machine Learning Startup *Co-Founder*
  - Implemented **PixelLink** and a GRU for word-level text detection, **invariant** to font size, colour, background, orientation, etc. and demonstrated an accuracy of **74%** on a proprietary dataset of documents like Aadhar Card, Passport, Driving Licence, etc.
  - Implemented a robust model with a **UNet** backbone for **segmenting** out spots, patches and wrinkles in selfies and exposed it through **AWS for demonstration**

Research Experience

- **Switching Subspace Model for Neural Population Analysis** *Sep'22 - May'24*
  Guide: Prof. Anqi Wu, GeorgiaTech *Research project*
  - Developed a sequential VAE model to capture multiple latent dynamics throughout a trial
  - Introduced unimodal Gaussian Process prior to improve interpretability
  - Introduce supervision using CNNs in order to preserve behavioral variable of interest
- **Assessing Comprehensibility of Children's Read Speech** *May'21 - June'22*
  Guide: Prof. Preeti Rao, IIT Bombay | arXiv:2405.19426 *Master's Thesis II*
  - Implemented a Wav2vec2.0-based end-to-end model for oral reading fluency assessment
  - Outperformed RFC baseline operating on hand-crafted features by 0.06 (absolute Pearson)
  - Probed the internal representations for presence of knowledge-based features using MLP probes
  - Implemented a multi-task learning framework for exploiting hand-crafted features during training
  - Proposed a self-supervised learning framework for utilising an unlabelled dataset
- **Prominence Detection in Children's Read Speech** *Jan'21 - Oct'21*
  Guide: Prof. Preeti Rao, IIT Bombay | arXiv:2110.14273 *Master's Thesis I*
  - Replaced a Random Forest Classifier baseline with a **CRNN** framework for predicting the degree of prominence for each word in children's read speech
  - Explored inputs at various hierarchies: raw waveforms, acoustic contours and word-level aggregates
  - Demonstrated an improvement in the acoustic features extracted from word segments using Sinc convolution
  - Exploited phrase boundary labels in various **multi-task learning** paradigms
  - Used part-of-speech tags and various **NLP** embeddings such as **GloVe** and **BERT** for incorporating complementary lexical information
- **Character Animation from Video for Blender** *July'21 - Dec'21*
  Guide: Prof. Parag Chaudhuri, IIT Bombay *Research Project*
  - Working on a Blender plugin consisting of an **integrated pipeline** for extracting **3D human pose** from a video using various deep learning backends and **retargeting** it to a rigged character in Blender
  - Added VIBE and MediaPipe to the pose extraction backend of the plugin
  - Explored a **self-supervised** graph neural network framework for dynamic mapping of animations from source to dissimilar target skeletons
- **SIRD Dynamics** *Aug'20 - Dec'20*
  Guide: Prof. Sharayu Moharir, IIT Bombay *Research Project*
  - Studied the **SIRD** model which is widely used for studying the outbreak of epidemics
  - Simulated the model with various underlying **network topologies** in place of random mixing
  - Formulated multiple mathematical models for calculation of **precise dynamics**

Key Projects

- **WhichLORA** | Course Project: Systems for AI: LLMs *Jan'24 - Apr'24*
  - Developed an end-to-end two-stage framework for automatic generation of LoRA variants based on user constraints for fine-tuning and inference compute and latency
  - Profiled various hardware configurations such as NVIDIA A100, H100 and RTX Quadro 6000
  - Developed a memory predictor for predicting HBM usage during inference given a LLM configuration
- **Causal Relationships in Diabetes Prediction** | Course Project: Machine Learning *Aug'23 - Dec'23*
  - Conducted EDA of the 21 features using violin plots, heatmaps and bar charts
  - Implemented various baselines such as SVM, Random Forests and XGBoost for predicting diabetes
  - Uncovered various causal relationships by designing various DAGs
- **Compiler Optimizations for speeding up Capsules** | Course Project: Parallelizing Compilers *Mar'23 - May'23*
  - Implemented compiler optimization techniques such as unroll and jam, scalar replacement and loop reordering in CUDA for reducing execution time of the core capsule operation
  - Explained the **4x speedup** in CUDA execution time by examining the profiler output
- **Legendre Memory Unit** | Course Project: Advanced Machine Learning *Jan'21 - May'21*
  - Implemented and analysed the performance of Legendre Memory Units (LMU), **an improved sequential model**, on various tasks and datasets such as JSB Chorales, Mackey-Glass dynamics, etc.
  - Suggested modifications to the **core equations** by studying various **basis** functions
- **Audio Steganography** | Course Project: Automatic Speech Recognition *Jan'21 - May'21*
  - Exploited **adversarial attacks** on ASR systems for hiding any given sequence of tokens in any audio file
  - Analysed performance as a function of various token sequence properties such as **length and perplexity**
  - Demonstrated **high PESQ scores** which indicate low perceptibility of deviation from original audio
- **Video Toonification** | Course Project: Digital Image Processing *Aug'20 - Dec'20*
  - Used **Mean Shift Segmentation** across both time and spatial dimensions for toonification of videos
  - Benchmarked results with standard techniques such as Bilateral Filtering
- **Auction Theory** | Course Project: Game Theory *Aug'20 - Dec'20*
  - Studied various models in Auction Theory such as **Vickrey Auction** and **First Price Sealed Bid** Auctions
  - Discussed equilibrium and optimal auction design analysis
- **FMX Rendering and Animation** | Course Project: Computer Graphics *Nov'20 - Dec'20*
  - Designed and rendered an FMX track with obstacles of varying shapes such as **cylinders and ramps**
  - Designed, rendered and animated a rider and a motorbike on the track using **keyframing**
  - Employed **Phong Shading, Texture Mapping** and used a **Skybox** for a realistic look
- **Pipelined RISC Processor** | Course Project: Microprocessors *Oct'19 - Nov'19*
  - Designed a 16-bit, 8-register, 6-stage **Pipelined** RISC processor in VHDL
  - Employed **Branch Prediction** and **Hazard Mitigation** techniques for optimizing the performance
- **FindIt** | Self Project: Audio Fingerprinting *May'19 - June'19*
  - FindIt is a Python program for identifying a song given a **short noisy segment**, similar to Shazam
  - An **audio fingerprint** consisting of constellations of major time-frequency peaks is stored in a hash table
- **Handwriting Recognition Pen** | Summer Project *May'18 - June'18*
  - Built a pen which can instantly convert **handwriting strokes** on ordinary paper into text
  - Designed the pen from scratch in AutoCAD and **3D printed** it
  - Generated own training data for each letter using a custom OpenCV script
- **Furniture Classification** | Kaggle Competition *May'18 - June'18*
  - Participated in the iMaterialist Challenge organised by Malong Technologies and CVPR 2018
  - Implemented **ResNet in PyTorch** for classification of furniture images into **128 classes**, each class containing around 1500 training images with low inter-class variation
  - Achieved an accuracy of 87.4% and ranked **30** among 428 teams across the globe

Technical Skills

- **Programming Languages:** Python, C++, C, Bash, Verilog, VHDL, OpenGL, SQLite
- **Softwares:** Matlab, Arduino, LaTeX, Blender, VHDL, AutoCad, Solidworks, Android Studio
- **Data Science:** PyTorch, Pandas, Numpy, OpenCV, TensorFlow, MATLAB

Key Coursework

- **Computer Science:** Logic, Compilers, Systems for AI, Game AI, Automatic Speech Recognition, Advanced Machine Learning, Computer Graphics, Foundations of Intelligent and Learning Agents, Data Structures and Algorithms, Digital Image Processing, Network Security, Computer Networks, Operating Systems, Data and Visual Analysis
- **Electrical Engineering:** Digital Communication, Digital Systems, Digital Signal Processing, Data Analysis and Interpretation, Control Systems, Information Theory and Coding, Markov Chains, Microprocessors, Microelectronics, Foundations of VLSI CAD, Power Systems
- **Miscellaneous:** Non-linear Optimization, Dynamic Algebraic Algorithms, Calculus, Complex Analysis, Linear Algebra, Differential Equations, Biology, Chemistry, Economics, Psychology, Engineering Drawing, Environmental Studies

Positions of Responsibility

- **Editorial Board Member, Insight** *Apr'21 - Mar'22*
  Insight is IIT Bombay's student media body with over 10,000+ readers
  - Surveyed the **effectiveness** of the Faculty Advisor program by taking inputs from both students and faculty and **suggested various reforms**
  - Initiated a series on **startups from research labs** at IIT Bombay as part of the **LinkedIn** team
  - Interviewed authorities and current international students for understanding the causes behind **poor international representation** at IITB and suggested **remedies** for the same
  - Awarded **Special Mention for Journalism** by IIT Bombay for work carried out as part of Insight
- **Department Academic Mentor** *Apr'21 - June'22*
  - Selected as part of a 35-member team on the basis of **ethics, peer-reviews** and an **interview**
  - **Mentoring** 6 sophomores in academic and co-curricular activities
- **Teaching Assistant** *July'21 - Apr'22*
  - TA for EE679, a graduate-level course on Speech Processing which covers speech production, analysis techniques and applications such as ASR, speech synthesis, etc. (Autumn 2021-22)
  - TA for EE352, a lab course on Digital Signal Processing which covers practical aspects of signal processing algorithms such as FFT on digital signal processors (Spring 2021-22)

Extra-Curricular Activities

- National-level quarter-finalist at Bournvita Quiz Contest; appeared on **National TV** for the same
- **Won 2nd prize** in **Android app development** competition organised by Web and Coding Club
- Successfully completed a 12-month **Lawn Tennis** course under National Sports Organisation and represented Hostel 4 in inter-hostel tournaments
- Attended the **Vijyoshi** camp conducted by **IISC**, Bangalore which served as a platform for interaction between bright young students and leading researchers in the field of science
- Awarded **Best Outgoing Student** of the year 2014-15 by Nirmala Convent High School